TON THAT GIA HUY

Data Engineering & Backend

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PROFILE

Data Engineering-focused Data Science candidate with project experience building Python and SQL ingestion workflows, PostgreSQL lifecycle controls, multi-format document processing, and PySpark batch pipelines. Builds traceable data flows and backend APIs using automated tests, Docker, and Git-based workflows.

EDUCATION

University of Transport Ho Chi Minh City (UTH) 2023-2027
Bachelor of Data Science

PROJECTS

CloudMentor AI - Document Ingestion & Lifecycle

Personal project | Python, FastAPI, PostgreSQL, Chroma, MinIO, Docker | GitHub

- Built PDF, TXT, and DOCX ingestion with page, line, or paragraph lineage through chunking and Chroma indexing; PostgreSQL manages document identity and lifecycle state.
- Implemented guarded local-to-MinIO migration and recovery, database-backed duplicate protection, and bounded PostgreSQL-Chroma reconciliation with automated CI checks.

OpsBrain - Versioned Data Contract & Ingestion

Team learning project | Data Engineering/Backend scope | Python, PostgreSQL, SQLAlchemy | GitHub

- Implemented a versioned Markdown frontmatter contract and Pydantic/PyYAML validator covering shared fields, six document types, and relationship checks.
- Built Alembic/SQLAlchemy migrations and PostgreSQL flows for create, unchanged replay, and immutable semantic updates using canonical content hashes.

Social Media Sentiment Big Data Pipeline

Coursework project | Python, PySpark, HDFS, Parquet, MongoDB, FastAPI | GitHub

- Processed approximately 1.6 million Sentiment140 records through HDFS and PySpark cleaning, feature generation, and partitioned Parquet outputs.
- Exposed saved analytics through FastAPI and MongoDB; built a separate Kafka demo using synthetic messages for event-ingestion practice.

TECHNICAL SKILLS

Core: Python, SQL, ETL, PostgreSQL
Data Processing: Pandas, PySpark, Parquet, HDFS, Kafka fundamentals
Backend & Delivery: FastAPI, Pydantic, SQLAlchemy, Alembic, Chroma, MinIO, Docker, Git, pytest, GitHub Actions

CERTIFICATION

Databricks Fundamentals Accreditation - Databricks Academy, Sep 2026

LANGUAGES

English: B2 (CEFR, self-reported)